SafeRoad AI

Road Accident Severity Prediction Using Machine Learning

PROJECT REPORT

A machine learning-based web application for predicting road accident severity from environmental and traffic-related conditions.

Python • Pandas • NumPy • Scikit-learn • Random Forest • Streamlit

1. Abstract

SafeRoad AI is a machine learning project designed to predict the severity of road accidents using information about accident and traffic conditions. The system uses a Random Forest classifier trained on road accident data. The data is cleaned and prepared before training, and categorical features are converted into numerical values using label encoding. The trained model is saved and connected to an interactive Streamlit web application. Users can enter conditions such as weather, road surface, lighting, road type, speed limit, number of vehicles, and number of casualties to obtain a predicted accident severity. During model development, the reported test accuracy was approximately 85.17%. This value is based on the particular test split used during development and should not be treated as a guarantee of real-world prediction accuracy.

2. Introduction

Road accidents are influenced by many factors, including weather, road conditions, lighting, vehicle count, speed, and other traffic circumstances. Understanding these factors can help in studying accident patterns and building data-driven prediction systems. SafeRoad AI applies machine learning to this problem. The system learns patterns from historical accident data and uses those patterns to estimate the severity category for new inputs. A Streamlit interface makes the trained model easy to test without writing Python code.

3. Problem Statement

Road accident datasets can contain many records and several factors that affect accident outcomes. Manually analyzing these relationships can be difficult and time-consuming. SafeRoad AI aims to develop a simple machine learning system that learns from historical accident data and predicts the likely severity of an accident based on selected input conditions.

4. Objectives

• Collect and work with road accident data.
• Clean and preprocess the dataset for machine learning.
• Explore accident patterns using data analysis and visualizations.
• Convert categorical data into numerical form.
• Train a Random Forest classification model.
• Evaluate the model using a test dataset.
• Save the trained model and encoders.
• Build an interactive Streamlit application for predictions.
• Organize the project into a clear GitHub-ready structure.

5. Dataset Description

The project uses a road accident dataset containing information about accident conditions and outcomes. The original CSV is kept for reference, while a cleaned CSV is maintained for analysis and model development. The dataset contains environmental, road, traffic, and accident-related information.

6. Input Features

The application uses the following features: Weather Conditions, Road Surface Condition, Light Condition, Road Type, Speed Limit, Number of Vehicles, and Number of Casualties. These inputs represent environmental and traffic conditions that can be supplied to the trained classifier.

7. Data Preprocessing

Before model training, the data is prepared so it can be processed by the machine learning algorithm. The workflow includes checking and cleaning the dataset, selecting useful features, and converting categorical values into numerical values. LabelEncoder from scikit-learn is used for categorical columns. The encoders are saved using Joblib so the same transformations can be applied to new inputs in the Streamlit application.

8. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the structure and patterns in the accident dataset before building the model. The EDA notebook examines the data and uses charts to study accident severity and relationships with weather, road conditions, lighting, and traffic conditions. These visualizations help identify patterns and support better understanding of the dataset.

9. Methodology

The project follows an end-to-end workflow: Data Collection → Data Cleaning → Exploratory Data Analysis → Feature Encoding → Model Training → Model Evaluation → Model Saving → Streamlit Application. This workflow connects data science and machine learning with a usable web interface.

10. Machine Learning Model

The project uses the Random Forest classifier. Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to make a classification decision. It can learn relationships between multiple input features and the target severity category. The trained model is stored as models/accident_model.pkl and the categorical encoders are stored as models/encoders.pkl.

11. Model Performance

Reported test accuracy: approximately 85.17%. The reported accuracy comes from the test split used during model development. It indicates how the model performed on that particular held-out data. It should not be interpreted as a guarantee of real-world prediction accuracy. Additional validation and independent testing would be useful before real-world use.

12. Web Application

The prediction interface is developed using Streamlit. The user enters road and accident conditions through interactive controls. The values are transformed using the saved encoders and passed to the saved Random Forest model. The application then displays the predicted accident severity. The main application file is app/app.py.

13. Project Structure

SafeRoad-AI/
■■■ app/app.py
■■■ data/Road Accident Data.csv
■■■ data/cleaned_accident_data.csv
■■■ images/
■■■ models/accident_model.pkl
■■■ models/encoders.pkl
■■■ notebooks/EDA.ipynb
■■■ reports/SafeRoad_AI_Project_Report.pdf
■■■ requirements.txt
■■■ README.md

14. Technologies Used

Python, Pandas, NumPy, Matplotlib, Scikit-learn, Random Forest, Joblib, Streamlit, Jupyter Notebook, Git, and GitHub are used across the project.

15. Results

The completed project provides an end-to-end machine learning workflow. Accident data is prepared and explored, a Random Forest classifier is trained, the model and encoders are saved, and a Streamlit interface is used to make predictions. The model achieved approximately 85.17% test accuracy on the development test split. The GitHub repository contains the main project components, including the application, data, trained model, encoders, notebook, and README.

16. Limitations

• The model depends on the quality and representativeness of the available dataset.
• Test accuracy alone does not fully describe classification performance.
• The prediction should be considered an analytical output rather than a real-world safety guarantee.
• More independent data and additional validation would improve confidence in the model.

17. Future Scope

• Add more recent and geographically diverse accident data.
• Compare Random Forest with other classification algorithms.
• Include precision, recall, F1-score, and confusion matrix analysis.
• Improve the user interface with more visual explanations.
• Add model monitoring and periodic retraining.
• Deploy the application online for demonstration and portfolio use.

18. Conclusion

SafeRoad AI demonstrates how machine learning can be used to analyze road accident data and predict accident severity from selected conditions. The project combines data preprocessing, exploratory analysis, Random Forest classification, model saving, and Streamlit application development. It provides a practical example of an end-to-end data science and machine learning project. Further testing and validation would be required for real-world use.

19. References

1. Scikit-learn documentation – Machine Learning in Python.
2. Pandas documentation – Data analysis and manipulation in Python.
3. NumPy documentation – Numerical computing with Python.
4. Matplotlib documentation – Visualization with Python.
5. Streamlit documentation – Python framework for data applications.
6. Joblib documentation – Persistence utilities for Python objects.

Keywords: Road Accident Analysis, Machine Learning, Random Forest, Accident Severity, Python, Streamlit, Data Science